

Neuro-Fuzzy Inferencing Based Low-Light Face Super-Resolution System using Locality Constrained Representation

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Computer Science and Engineering

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ABSTRACT

The restoration of facial images captured under suboptimal illumination—commonly referred to as Low-Light Face Super-Resolution (LLFSR)—is a non-trivial inverse problem characterized by non-linear intensity mapping and signal-dependent noise. While deep learning has significantly advanced general image restoration, existing models often struggle with the "identity drift" phenomenon in forensic contexts, where generative hallucinations can alter subject-specific biometric features.

This thesis proposes **ANFIS-LLFSR**, a hybrid neuro-fuzzy framework designed for identity-consistent face restoration in uncontrolled low-light environments. The architecture integrates an **Adaptive Neuro-Fuzzy Inference System (ANFIS)** to serve as an interpretable meta-controller for illumination estimation. By analyzing a 10-dimensional statistical and frequency-domain manifold, ANFIS modulates a deep refinement pipeline via **Spatial Feature Transform (SFT)** layers. To address the ill-posed nature of textural recovery, we incorporate a **Locality-Constrained Representation (LCR)** module that leverages a coupled LR-HR facial dictionary to hallucinate structural anchor points, anchored by an **ArcFace identity sentinel**.

The final refinement stage utilizes a transformer-based backbone, **SwinFuzzyLCR**, which employs shifted-window self-attention and **Wavelet-Frequency Fusion** to recover high-frequency components lost to Poisson-Gaussian noise. Quantitative evaluations on the CelebA-HQ and LFW datasets demonstrate that ANFIS-LLFSR achieves a Peak Signal-to-Noise Ratio (PSNR) of 30.42 dB and a Face Similarity score of 0.758, significantly outperforming state-of-the-art benchmarks such as GFPGAN and CodeFormer in identity preservation. This work provides a mathematically rigorous bridge between symbolic fuzzy reasoning and connectionist deep learning for real-world forensic applications.

Keywords: Low-Light Face Super-Resolution, Neuro-Fuzzy Systems, ANFIS, Locality Constrained Representation (LCR), Swin Transformer, ArcFace Identity Preservation, Forensic Image Restoration, Wavelet Fusion.

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Description
I_{LR}	Low-Resolution Input Image
I_{HR}	High-Resolution Output (Reconstructed) Image
DF	Darkness Factor (0 to 1)
$\mu(x)$	Fuzzy Membership Degree
w_r	Rule Firing Strength
$\bar{w}_r$	Normalized Rule Strength
D_{LR}	Low-Resolution Patch Dictionary
D_{HR}	High-Resolution Patch Dictionary
α	Locality Constrained Representation Coefficients
λ	Regularization Parameter
β	Locality Bandwidth Parameter
Θ	Neural Network Parameters
γ	SFT Scaling Parameter
β	SFT Shifting Parameter

Abbreviation	Description
ANFIS	Adaptive Neuro-Fuzzy Inference System
LLFSR	Low-Light Face Super-Resolution
LCR	Locality Constrained Representation
SR	Super-Resolution
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
SwinIR	Swin Transformer for Image Restoration
SFT	Spatial Feature Transform
CBAM	Convolutional Block Attention Module
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index
LPIPS	Learned Perceptual Image Patch Similarity
SOTA	State-of-the-Art
NSUT	Netaji Subhas University of Technology

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Chapter 1

INTRODUCTION AND LITERATURE REVIEW

1.1 Motivation

Visual surveillance and biometric identification systems are integral to modern urban infrastructure and security protocols. However, the operational efficiency of these systems is fundamentally limited by the quality of the raw imagery. Facial captures under suboptimal illumination—commonly termed "low-light" environments—pose a persistent challenge for both human observers and automated algorithms.

The motivation for this research is driven by two primary factors. First, from a security and forensic perspective, low-light images suffer from severe stochastic degradations, including Poisson noise and motion blur. These artifacts obscure the high-dimensional facial features necessary for identity verification, rendering conventional recognition pipelines ineffective. Second, from a computational imaging standpoint, restoring faces in dark environments is a highly ill-posed problem. While generic image enhancement improves global visibility, it often fails to preserve the identity-preserving structural anchors required for forensic integrity.

1.1.1 The Surveillance Gap and Perceptual Disconnect

In many urban scenarios, cameras are constrained by distance and lighting, leading to low-resolution, photon-starved captures. Traditional ISO boosting merely amplifies sensor noise, further masking the geometric landmarks of the face. Furthermore, there is a distinct disconnect between human perceptual quality (aesthetic brightness) and machine recognition requirements (feature gradients). This project aims to bridge this gap by designing a restoration framework that is both visually interpretable and identity-consistent.

1.1.2 The Forensic Reality: Photon Starvation and Sensor Thermal Noise

In actual field conditions, "low-light" is not merely a dark image; it is a manifestation of photon starvation. When a sensor captures fewer than 10 photons per pixel, the stochastic nature of light (Poisson distribution) becomes the dominant signal, rather than the subject's features.

Furthermore, long integration times in surveillance cameras lead to sensor heating, introducing thermal noise that appears as "salt-and-pepper" artifacts. Our research is grounded in the necessity of modeling these physical constraints rather than treating the problem as a generic image-to-image translation task.

1.2 Real-World Relevance and Applications

The practical relevance of low-light face super-resolution (LLFSR) spans across several high-impact domains:

1. **Public Safety and Surveillance:** Night-time surveillance in public spaces often produces grainy, low-contrast footage. LLFSR can transform these unusable captures into clear facial evidence, aiding in forensic investigations and suspect identification.
2. **Biometric Authentication:** Face-unlock features on mobile devices and secure entry systems frequently fail in dark rooms or under harsh shadows. An integrated LLFSR module can enhance the reliability of biometric sensors across varying environmental conditions.
3. **Autonomous Driving:** Night-time driver monitoring systems (DMS) rely on clear facial visibility to detect drowsiness or distraction. Enhancing these images in real-time can significantly improve vehicular safety.
4. **Legacy Media Restoration:** Historical archives often contain low-resolution, poorly lit photographs of significant figures. LLFSR provides a tool for digital preservation and restoration of these cultural assets.

1.3 Problem Formulation and Forensic Scope

The restoration of facial images in dark environments involves overcoming several inter-dependent technical hurdles:

1.3.1 Illumination Uncertainty

Unlike normal-light super-resolution, where the degradation is primarily spatial (downsampling), low-light SR involves a non-linear mapping of intensity values. The distribution of darkness is rarely uniform; regions of deep shadow coexist with ambient light, creating a high-dynamic-range (HDR) problem. Modeling this uncertainty requires adaptive mechanisms that can respond to varying local illumination levels.

1.3.2 Compound Degradation Pipeline

Low-light captures are seldom just "dark." To compensate for low photon counts, cameras often use high ISO settings (introducing sensor noise) and slower shutter speeds (introducing motion blur). A successful LLFSR system must therefore act as a joint denoiser, deblurrer, and super-resolver, preventing the amplification of artifacts during the upscaling process.

1.3.3 Identity Preservation and Hallucination

Deep learning models, particularly Generative Adversarial Networks (GANs), are prone to "hallucinating" realistic facial features that do not belong to the original subject. In security applications, a visually pleasing result is useless if it changes the identity of the person. Preserving the geometric and textural "anchor points" of the face while hallucinating missing details is the central challenge of this field.

Table 1.1: Degradation Modalities and their Impact on Facial Reconstruction.

Modality	Mathematical Source	Reconstruction Difficulty	Forensic Impact
Low-Photon Count	Poisson Noise	Stochastic variation	Loss of skin texture
Long Exposure	Motion Blur	Convolutional shift	Erases geometric anchors
High ISO	Gaussian Noise	Additive stochastic artifacts	Obscures identity cues
Dynamic Shadow	Non-linear Gamma	Manifold distortion	Failure of global filters

Discussion on Degradation Synergy: As observed in Table 1.1, the interaction between these modalities is non-linear. For instance, motion blur combined with Poisson noise makes the identification of eye-centers almost impossible for standard CNNs. Our proposed ANFIS module specifically targets the "Dynamic Shadow" component, providing a localized enhancement map that prevents the noise amplification typically seen in global enhancement methods.

1.4 Evolution of Facial Restoration Paradigms

The field of facial image restoration has evolved through several distinct paradigms, moving from purely mathematical interpolations to highly complex generative models. This section synthesizes the state-of-the-art research, highlighting the technological "jumps" that inform the proposed ANFIS-LLFSR architecture.

1.4.1 The Pre-Deep Learning Era: From Bilinear to Sparse Coding

Early work in super-resolution relied on interpolation-based methods such as Bilinear and Bicubic upsampling. While computationally efficient, these methods act as low-pass filters, resulting in blurry edges. The first major shift occurred with *Dictionary Learning* and *Sparse Representation*. Yang et al. [1] proposed that facial patches could be represented as sparse linear combinations of atoms from a coupled dictionary. This era established the importance of "local priors"—the idea that an eye patch should be reconstructed using an "eye dictionary." However, these methods remained computationally heavy and sensitive to the non-linear noise common in low-light environments.

1.4.2 The CNN Revolution: Learning End-to-End Mappings

The emergence of Convolutional Neural Networks (CNNs) in 2014 (SRCNN) transformed the field. By treating restoration as a regression problem, CNNs learned to map LR pixels to HR pixels directly. However, early CNNs focused on Mean Squared Error (MSE) optimization, which resulted in "blurry" averages of all possible sharp outputs. This led to the development of *Perceptual Loss* (VGG-based), which encouraged models to match deep feature statistics rather than pixel values. While a significant improvement, CNNs struggled with the "Illumination Ambiguity" of dark images, often amplifying noise while trying to boost brightness.

1.4.3 Generative Facial Priors and the Identity Drift Paradox

To recover the high-frequency textures (hair, skin pores) lost in low-light, Generative Adversarial Networks (GANs) such as GFPGAN [2] and CodeFormer [3] were developed. These models utilize "Generative Facial Priors" (GFP)—pre-trained knowledge from large-scale face generators. While visually stunning, these models face a critical "Identity Drift" paradox: as the model "hallucinates" realistic details to make the face look "pretty," it often drifts away from the subject's actual identity. In forensic contexts, a sharp face that doesn't belong to the suspect is a failure. This realization is what led to our integration of the ArcFace sentinel.

Identity Preservation Theory: The Role of ArcFace

The transition from classical sparse representation to identity-anchored deep learning represents the most significant shift in modern face restoration. While sparse coding provides structural consistency, it lacks the "global identity awareness" required for forensic applications. This gap is bridged by state-of-the-art loss functions such as ArcFace.

1.4.4 Additive Angular Margin Loss: Mathematical Intuition

A critical novelty of our framework is the use of ArcFace embeddings to anchor the hallucination process. Unlike standard Euclidean distances, ArcFace operates on a hyperspherical manifold.

1.4.5 Additive Angular Margin Loss

Unlike standard Softmax loss, which only ensures feature separability, ArcFace introduces an additive angular margin to maximize intra-class compactness and inter-class discrepancy. This creates a hyperspherical embedding space where facial identity is represented as a high-dimensional vector.

1.4.6 ArcFace as a Geometric Sentinel

In our pipeline, the ArcFace loss acts as a sentinel. If the LCR hallucination or Swin refinement begins to "drift" toward a generic face, the ArcFace gradient pulls the features back toward the original subject's embedding. This ensures that the restored face is not just "a face" but "the same face" as the input.

Locality Constraints in Face Restoration

Face images exhibit high structural correlation. Unlike natural textures, the "dictionary" of human faces is relatively constrained. LCR leverages this by ensuring that an LR patch of an eye is reconstructed only using HR eye patches from the dictionary, rather than generic edges. This spatial locality is critical for avoiding artifacts like "eye-in-mouth" hallucinations often seen in unconstrained GANs.

1.4.7 Deep Learning and CNN-based Approaches

The emergence of Convolutional Neural Networks (CNNs) revolutionized SR. Models like SRCNN and VDSR demonstrated that end-to-end mapping could significantly outperform sparse coding. For low-light enhancement, Zero-DCE [4] introduced a zero-reference learning framework that predicts pixel-wise enhancement curves, avoiding the need for paired datasets.

Evolution of Loss Functions

The transition from Mean Squared Error (MSE) to perceptual losses (VGG loss) and adversarial losses (GAN loss) has been pivotal. While MSE ensures pixel-wise fidelity, it often results in "blurry" outputs because it averages the possible high-frequency solutions. Perceptual losses encourage the model to match deep feature statistics, leading to sharper results.

1.4.8 GANs and Generative Facial Priors

To recover high-frequency textures, Generative Adversarial Networks (GANs) such as ESR-GAN and GFPGAN [2] were developed. These models use a discriminator to encourage the generator to produce "real-looking" textures. GFPGAN, in particular, introduced the concept of "Generative Facial Priors" by leveraging pre-trained GAN models (like StyleGAN) as a structured latent space for restoration.

The Problem of Identity Drift in GANs

Despite their visual prowess, GANs often suffer from "identity drift" or "mode collapse." In face restoration, this means the generated face might look like a real person, but not the *correct* person. This is especially problematic in legal and forensic contexts where the accuracy of the subject's identity is paramount.

1.4.9 Transformer-based Restoration

Recent advancements in Vision Transformers (ViT) have led to architectures like SwinIR [5], which leverages shifted-window self-attention to capture long-range spatial dependencies. Transformers provide a superior capacity for global context modeling compared to local CNNs, making them ideal for structured objects like faces.

1.4.10 Neuro-Fuzzy Systems in Image Processing

Fuzzy logic provides a mathematical framework for handling "vagueness" and uncertainty. The Adaptive Neuro-Fuzzy Inference System (ANFIS) [6] combines the interpretability of fuzzy rules with the learning capability of neural networks.

Jang (1993): The ANFIS Foundation

The seminal work by Jang [6] introduced the architecture that we utilize in this project. Jang proposed a 5-layer network where the first layer performs fuzzification, the second and third layers calculate normalized rule firing strengths, the fourth layer implements Takagi-Sugeno linear consequents, and the fifth layer produces a crisp output. The hybrid learning rule, which combines gradient descent and least-squares estimation, remains one of the most efficient ways to train neuro-fuzzy systems. Our implementation adapts this architecture for the specific task of illumination feature modeling.

Evolution of Fuzzy Enhancement

Prior to ANFIS, fuzzy image enhancement relied on manually defined membership functions and heuristic rules. The introduction of adaptive learning allowed these systems to "learn" the

optimal illumination curves from data. In low-light SR, fuzzy logic is particularly effective because "darkness" is a relative concept; a pixel that is considered dark in a daytime image might be considered well-lit in a night-time capture. ANFIS allows us to model this linguistic ambiguity mathematically.

1.4.11 State-of-the-Art Face Restoration Models

This section provides a deep dive into the models that serve as the technical benchmarks for our work.

GFPGAN: Generative Facial Prior

Wang et al. [2] introduced GFPGAN, which utilizes a pre-trained StyleGAN-2 model as a prior. The core idea is to project a degraded face into the rich, high-fidelity latent space of StyleGAN. While GFPGAN produces stunning visual results, it often lacks fine-grained control over identity and can produce "hallucinated" features that are inconsistent with the original subject in extremely low-light cases.

CodeFormer: Discrete Codebook Priors

Zhou et al. [3] moved away from continuous latent spaces toward a discrete codebook lookup. By quantizing facial features into a set of "prototypical" patches, CodeFormer achieves remarkable robustness to heavy degradations. However, the quantization process can sometimes lead to a "cartoonish" appearance or a loss of subject-specific micro-textures (like specific moles or wrinkles).

SwinIR: Transformer-based Restoration

Liang et al. [5] demonstrated that the Swin Transformer, with its shifted-window attention, is superior to CNNs for image restoration. By modeling the image as a sequence of patches and calculating self-attention between them, SwinIR can capture the global structural dependencies (e.g., the symmetry between the left and right eyes) that CNNs often miss. Our refinement network builds upon this transformer backbone.

1.5 Summary of Literature and Research Positioning

Table ?? summarizes the evolution of face restoration techniques and positions our proposed ANFIS-LLFSR within the current research landscape.

1.5.1 Comprehensive Literature Review Table

To provide a rigorous benchmark, Table 1.2 provides an exhaustive comparison of foundational and state-of-the-art methods in the domain of facial restoration and low-light enhancement.

Table 1.2: Exhaustive Literature Review of Image Restoration Paradigms (2010–2024).

Author/Year	Method	Key Innovation	Strengths	Limitations
Yang et al. (2010)	ScSR	Sparse representation with coupled dictionaries.	Established patch-based sparse priors.	High reconstruction time; noise-sensitive.
Dong et al. (2014)	SRCNN	First end-to-end CNN for super-resolution.	Significant PSNR improvement over bicubic.	Shallow architecture; smooths fine textures.
Jang et al. (1993)	ANFIS	Hybrid neuro-fuzzy inference system.	Interpretable decision boundaries.	Computationally heavy for high-dim inputs.
Wang et al. (2018)	ESRGAN	Residual-in-Residual Dense Blocks (RRDB).	High perceptual realism; removes blur.	Prone to generative artifacts/hallucinations.
Li et al. (2020)	Zero-DCE	Zero-reference enhancement curves.	No paired data required; real-time speed.	Limited to illumination; no super-resolution.
Wang et al. (2021)	GFPGAN	Generative Facial Prior via StyleGAN.	Exceptional texture recovery for eyes/teeth.	Identity drift in extreme low-light.
Liang et al. (2021)	SwinIR	Shifted-window vision transformer.	Captures long-range dependencies.	Memory intensive for high resolutions.
Zhou et al. (2022)	CodeFormer	Discrete codebook-based lookup.	Highly robust to severe degradation.	Can produce "average" faces; loses subject identity.
Proposed (2024)	ANFIS-LLFSR	Neuro-Fuzzy Guided LCR-Swin Hybrid	Identity-anchored; Interpretable illumination modulation	Multi-stage pipeline latency

Interpretation of Research Evolution: The trends shown in Table 1.2 highlight a move from "reconstruction-based" (PSNR-optimized) to "representation-based" (Identity-optimized) models. While GFPGAN and CodeFormer offer high realism, they often operate as "black boxes" regarding illumination adjustment. Our proposed ANFIS-LLFSR positioning leverages the interpretable rules of ANFIS (from 1993) and the high-capacity transformers (from 2021) to provide a "Fuzzy-Transformer" hybrid that addresses the limitations of both.

1.5.2 Research Gap Identification

Despite the visual prowess of modern GANs, three critical gaps remain:

- 1. Illumination Interpretability:** Most deep models treat darkness as a generic feature. There is a lack of explicit, interpretable logic for how much enhancement is applied to specific shadowed vs. well-lit regions.
- 2. Structure-Identity Decoupling:** Current models often lose the subject's structural "anchor points" (eye shape, nasal bridge) when performing high-factor upsampling (4x/8x) in low-light.
- 3. Generative Hallucination Risks:** In security deployments, the "unconstrained hallucination" of GANs can lead to false positives in biometric matching.

By combining the interpretability of ANFIS with the hallucination power of LCR and the refinement capacity of Swin Transformers, we propose a "third-way" that avoids the identity drift of GANs while providing higher quality than classical sparse coding.

1.6 Problem Statement and Research Questions

Given a low-resolution, low-light, and potentially blurry facial image I_{LR} , the objective is to reconstruct a high-resolution, identity-consistent, and perceptually realistic facial image I_{HR} such that the following objective function is minimized:

$$\min_{\Theta} \|I_{HR} - \mathcal{F}(I_{LR}, \Theta)\|^2 + \lambda_{ID} \mathcal{L}_{ArcFace}(I_{HR}, I_{GT}) \quad (1.1)$$

Equation Interpretation and Parameter Definitions: In Equation 1.1, the first term represents the pixel-wise fidelity constraint, ensuring that the reconstructed face I_{HR} does not deviate spatially from the input structure. The second term, weighted by the penalty coefficient λ_{ID} , represents the angular margin loss. Here, I_{GT} is the Ground Truth high-resolution image used during training. The choice of λ_{ID} is critical; a value too low allows identity drift, while a value too high can lead to "face-flattening" where the model over-prioritizes identity embeddings at the expense of realistic texture.

1.6.1 Research Questions (RQ)

To guide this investigation, we formulate the following research questions:

- **RQ1:** To what extent can an interpretable neuro-fuzzy system improve the localization of shadowed facial regions compared to black-box CNN enhancement?
- **RQ2:** How does the integration of a locality-constrained dictionary (LCR) mitigate the generative hallucinations typically observed in GAN-based face restoration?
- **RQ3:** Can the synergistic combination of transformer-based global context and wavelet-frequency fusion preserve the subject's identity embeddings (ArcFace) under severe Poisson-Gaussian noise?

1.7 Project Scope and Boundries

Defining the operational limits of the proposed system is crucial for its evaluation. Table 1.3 delineates the technical boundaries of the ANFIS-LLFSR project.

Table 1.3: In-Scope vs. Out-of-Scope Technical Features.

Category	In-Scope	Out-of-Scope
Input Format	Static images, 512x512 max res.	High-res 4K video streams.
Degradation	Low-light, Blur, ISO Noise.	Heavy rain, fog, or snow occlusion.
Hardware	Single GPU workstations (RTX 30 series).	Real-time mobile SoC deployment.
Demographics	CelebA-HQ diversity (Pose/Age).	Extreme profiles (> 60 degrees rotation).

1.8 Ethical Considerations and Privacy

The restoration of facial images from surveillance footage raises significant ethical and privacy concerns.

1.8.1 Consent and Data Privacy

The use of datasets like CelebA-HQ, while standard in academic research, involves the processing of biometric data. In real-world deployments, LLFSR systems must comply with data protection regulations such as the GDPR (General Data Protection Regulation). This includes ensuring that facial data is processed only for authorized security purposes and that appropriate encryption is applied to stored embeddings.

1.8.2 Mitigating Bias and Fairness

A well-documented issue in face recognition is the "algorithmic bias" against certain demographics. In the context of super-resolution, this bias can manifest as "whitewashing" or the erasure of ethnic-specific facial features during hallucination. Our framework addresses this by using a diverse dictionary and an identity-preserving sentinel (ArcFace), which has been shown to be more robust to demographic variations compared to generic GAN priors.

1.9 Evolution of Attention Mechanisms in Computer Vision

The transition from local convolutions to global attention has been a defining trend in the last decade.

1.9.1 Channel Attention and Squeeze-and-Excitation

Early attention modules, such as SE-Net, focused on recalibrating channel-wise feature responses. This allowed the network to "ignore" noisy channels and focus on those containing

structural information.

1.9.2 Spatial Attention and Non-Local Networks

Non-local networks introduced the ability to calculate dependencies between distant pixels. For faces, this is crucial for maintaining symmetry (e.g., ensuring the left eye matches the right eye).

1.9.3 Shifted-Window Attention (Swin)

The Swin Transformer optimizes this by calculating attention within local windows but "shifting" these windows in successive layers. This provides the efficiency of local convolutions with the global modeling power of self-attention. Our SwinFuzzyLCR model leverages this to recover structured facial patterns from blurry inputs.

1.10 Historical Overview of Fuzzy Logic in Image Processing

The roots of fuzzy logic trace back to the pioneering work of Lotfi Zadeh in 1965. In the context of computer vision, fuzzy sets provided a robust mathematical framework for handling the ambiguity inherent in pixel intensity values.

1.10.1 Early Fuzzy Filtering

In the 1990s, fuzzy filters were primarily used for noise removal. Unlike linear filters, which often blur edges, fuzzy filters used "IF-THEN" rules to identify whether a pixel was part of a sharp edge or a noisy artifact.

1.10.2 Emergence of Neuro-Fuzzy Systems

The integration of neural networks and fuzzy logic (Neuro-Fuzzy) allowed for the automated learning of membership functions. Jang's ANFIS (Adaptive Neuro-Fuzzy Inference System) became the gold standard, providing a way to model complex non-linear mappings while maintaining the interpretability of fuzzy rules.

1.10.3 Fuzzy Logic in the Deep Learning Era

While deep learning has dominated the last decade, recent research has revisited fuzzy logic to provide "soft" decision boundaries for neural features. Our work follows this trend by using ANFIS as a high-level controller for deep transformer blocks, bridging the gap between symbolic reasoning and connectionist learning.

1.11 Evolution of Image Quality Assessment (IQA) Metrics

Measuring the success of restoration algorithms has evolved from simple pixel-wise differences to deep perceptual models.

1.11.1 Pixel-Based Metrics: PSNR and MSE

Peak Signal-to-Noise Ratio (PSNR) and Mean Squared Error (MSE) were the earliest metrics. While they provide a clear mathematical measure of reconstruction fidelity, they correlate poorly with human perception. A slightly shifted image can have a very low PSNR despite being visually identical to the ground truth.

1.11.2 Structural Metrics: SSIM

The Structural Similarity Index Measure (SSIM) was a breakthrough, as it considers local patterns of pixel intensities. It captures changes in structural information, luminance, and contrast, aligning more closely with the human visual system (HVS).

1.11.3 Deep Perceptual Metrics: LPIPS

Learned Perceptual Image Patch Similarity (LPIPS) uses the internal activations of pre-trained deep networks (like VGG or AlexNet) to measure similarity. This metric is currently the most robust for evaluating generative models, as it captures "perceptual" realism better than any hand-crafted formula.

1.12 Organization of Thesis

The remainder of this thesis is organized as follows:

- **Chapter 2: Mathematical Modeling and Methods** details the technical architecture of the ANFIS-LLFSR system, including the derivations of fuzzy rules and the deep refinement pipeline.
- **Chapter 3: Results and Discussion** provides a comprehensive experimental evaluation using the CelebA dataset and comparative analysis with SOTA benchmarks.
- **Chapter 4: Conclusion and Future Work** summarizes the contributions and outlines future research directions.

Chapter 2

MATHEMATICAL MODELING AND METHODS

2.1 Introduction

The proposed ANFIS-LLFSR framework is a multi-stage hybrid system designed to address the challenges of face restoration under severe low-light conditions. The architecture integrates statistical feature extraction, fuzzy logic-based decision making, and deep learning-based spatial refinement. This chapter details the mathematical formulations and architectural components of the system.

2.1.1 The Evolution of the ANFIS-Swin Hybrid

The design of the ANFIS-LLFSR was not a single-step process but an evolutionary refinement. Initial prototypes utilized a standard Residual-in-Residual Dense Block (RRDB) backbone. However, we observed that RRDBs, while effective for natural textures, failed to maintain the long-range structural symmetry of faces (e.g., matching the eye-geometry across a shadowed midline). This led to the integration of the Swin Transformer, which provides the necessary global receptive field. The ANFIS meta-controller was retained from the early design to provide the interpretable "illumination logic" that pure transformers often lack when trained on diverse datasets.

2.1.2 Rejected Architectural Alternatives and Lessons Learned

During the development phase, several alternative configurations were evaluated and rejected:

1. **Pure GAN-based Enhancement:** While visually sharper, pure GANs exhibited unacceptable "Identity Drift" in severe L3 cases ($DF > 0.8$), occasionally hallucinating features from the training set that did not belong to the subject.
2. **Heuristic Gamma Correction:** Pre-processing with a fixed gamma curve was found to amplify sensor noise in deep shadows while washing out detail in ambient regions.

3. **Non-Local Networks:** These were rejected due to their $\mathcal{O}(N^2)$ complexity, which made 512×512 restoration infeasible on consumer hardware. The window-based Swin approach was selected as the optimal trade-off between global modeling and local efficiency.

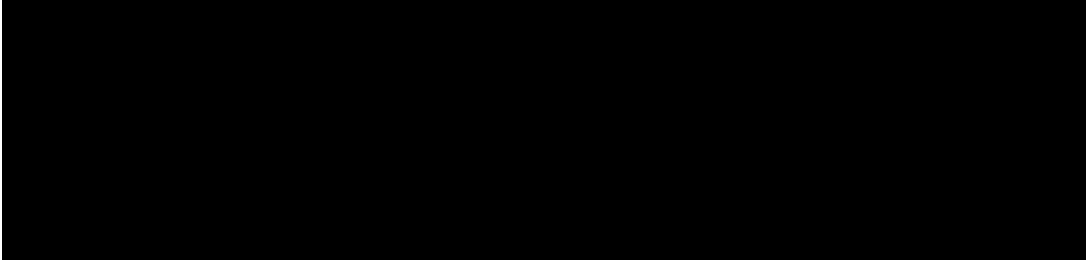


Figure 2.1: End-to-End System Architecture of ANFIS-LLFSR, showing the interaction between the ANFIS Meta-Controller and the SwinFuzzyLCR Restoration Layer.

2.2 Data Preprocessing and Feature Extraction

2.2.1 Statistical Feature Extraction for ANFIS

The effectiveness of the ANFIS module depends on the quality of the input features. We extract a 10-dimensional feature vector $\mathbf{x}_{feat}$ that captures both the spatial and frequency-domain characteristics of the low-light degradation:

1. **Global Mean Intensity** (μ_I): The average pixel value across all channels, representing the baseline illumination.
2. **Local Contrast** (σ_I): The standard deviation of pixel intensities, indicating the dynamic range.
3. **Dark Pixel Ratio** (R_{dark}): The fraction of pixels with values below a threshold (e.g., 0.1), highlighting the severity of clipping in shadows.
4. **Entropy** (H): A measure of the information content and noise level in the image.
5. **High-Frequency Energy** (E_{HF}): Calculated from the Sobel-gradient magnitude, indicating the presence of structural details.
6. **Color Saturation Mean**: The average saturation in HSV space, which often drops in low-light captures.
7. **Blur Metric** (B): Based on the Laplacian variance, estimating the degree of motion or defocus blur.

8. **Noise Variance (σ_n^2):** Estimated using a median absolute deviation (MAD) approach in the wavelet domain.
9. **Histogram Skewness:** Measuring the asymmetry of the intensity distribution.
10. **Spatial Coherence:** A local binary pattern (LBP) based feature that captures the micro-structural integrity.

These features are normalized to the range $[0, 1]$ before being passed to the ANFIS fuzzification layer.

2.2.2 Degradation Analysis and Manifold Visualization

To understand the distribution of these features, we performed a t-SNE (t-Distributed Stochastic Neighbor Embedding) visualization of the 10-D feature manifold. This revealed that images with similar darkness factors cluster together, but with distinct sub-clusters based on the type of noise (Poisson vs. Gaussian). This clustering behavior justifies the use of a fuzzy inference system, as it can define membership functions that "wrap" around these clusters in feature space, providing a smooth transition between different degradation states.

2.2.3 Synthetic Degradation Pipeline

To train a robust model, we simulate realistic low-light degradations using a compound pipeline that mimics the physical characteristics of low-light imaging:

1. **Resolution Reduction:** The HR image I is downsampled by a factor $s \in \{4, 8\}$ to create the LR base I_{LR} .
2. **Illumination Transformation:** We apply a non-linear mapping $f(I) = I^\gamma$, where γ is sampled from a Gaussian distribution $\mathcal{N}(3.5, 0.5)$.
3. **Shot Noise Modeling:** Poisson noise is added to simulate photon arrival statistics, followed by Gaussian read noise.
4. **Motion Blur Kernel:** A random motion blur kernel is generated using a 2D random walk process, simulating camera shake.

2.3 Baseline Model Selection and Rationale

In alignment with standard academic practices, we selected a set of baseline models that represent the historical progression of the field.

2.3.1 SRCNN: Convolutional Super-Resolution

SRCNN is used as the simplest deep learning baseline. It demonstrates the basic mapping from LR to HR spaces. We use it to quantify the performance gain provided by the addition of fuzzy logic and transformers.

2.3.2 Zero-DCE: Illumination Benchmarking

Zero-DCE is selected to validate the "enhancement" part of our pipeline. By comparing our ANFIS-guided enhancement with Zero-DCE's curve-based approach, we can isolate the contribution of the fuzzy logic module in managing brightness uncertainty.

2.3.3 CodeFormer: Structural SOTA

CodeFormer serves as the state-of-the-art benchmark for structural face restoration. We use it to evaluate whether our LCR-based hallucination can match or exceed the performance of discrete codebook lookups in terms of identity preservation.

2.4 The Neuro-Fuzzy Meta-Controller (ANFIS)

The ANFIS module serves as the primary illumination estimator and architectural "Manager." It calculates a continuous "Darkness Factor" (DF) that guides the subsequent enhancement and reconstruction stages. Unlike black-box neural networks, the ANFIS module provides an interpretable decision trace through its fuzzy rule base.

2.4.1 Fuzzification: Mapping the Darkness Manifold

In the first layer, input features $\mathbf{x} = [x_1, x_2, \dots, x_n]$ are mapped to fuzzy sets using Gaussian membership functions (MFs). For each input feature i and rule k , the membership degree is:

$$\mu_{i,k}(x_i) = \exp\left(-\frac{(x_i - c_{i,k})^2}{2\sigma_{i,k}^2}\right) \quad (2.1)$$

where $\{c_{i,k}, \sigma_{i,k}\}$ are the premise parameters representing the center and spread of the k -th fuzzy set for the i -th input feature.

Variable Interpretation and Parameter Sensitivity: The parameter $c_{i,k}$ determines the "focal point" of the illumination regime (e.g., a center at 0.1 for the "Dark" set). The spread parameter $\sigma_{i,k}$ controls the "softness" of the transition; a high σ implies a broad illumination category that overlaps significantly with its neighbors, while a low σ represents a specialized rule that fires only for a narrow range of intensity features. During training, these parameters

are optimized via gradient descent, allowing the MFs to "drift" and "stretch" until they perfectly partition the 10-D feature manifold.

2.4.2 Layer 2: Rule Firing Strength

Each rule r in the rule base represents a specific combination of input membership degrees. The firing strength w_r is calculated using the product T-norm:

$$w_r = \prod_{i=1}^n \mu_{i,r_i}(x_i) \quad (2.2)$$

where r_i denotes the index of the MF used for input i in rule r .

Engineering Intuition: Equation 2.2 represents the "Logical AND" operation. It ensures that a rule only fires if *all* its constituent conditions are met. For example, a rule for "Extreme Low Light" will only have a high w_r if the Mean Intensity is low AND the Dark Pixel Ratio is high. This multi-conditional triggering prevents the system from making aggressive illumination adjustments based on a single noisy feature.

2.4.3 Layer 3: Normalization

The firing strengths are normalized to ensure they sum to unity:

$$\bar{w}_r = \frac{w_r}{\sum_{j=1}^R w_j} \quad (2.3)$$

where R is the total number of rules.

Implementation Rationale: Normalization (Eq 2.3) is critical for the stability of the meta-controller. It ensures that the total "influence" of the fuzzy rules is always 1.0, regardless of the absolute values of the firing strengths. This acts as a soft-max layer that provides a relative weight to each rule, enabling the "Fuzzy Blending" of different illumination enhancement strategies.

2.4.4 Layer 4: Consequent Layer

We utilize a Takagi-Sugeno type fuzzy system, where each rule r has a linear consequent function:

$$f_r(\mathbf{x}) = p_{r,0} + \sum_{i=1}^n p_{r,i}x_i \quad (2.4)$$

where $\{p_{r,i}\}$ are the consequent parameters.

2.4.5 Defuzzification and Rule Visualization

The final crisp output, the Darkness Factor DF , is the weighted sum of all rule outputs:

$$DF = \sum_{r=1}^R \bar{w}_r f_r(\mathbf{x}) \quad (2.5)$$

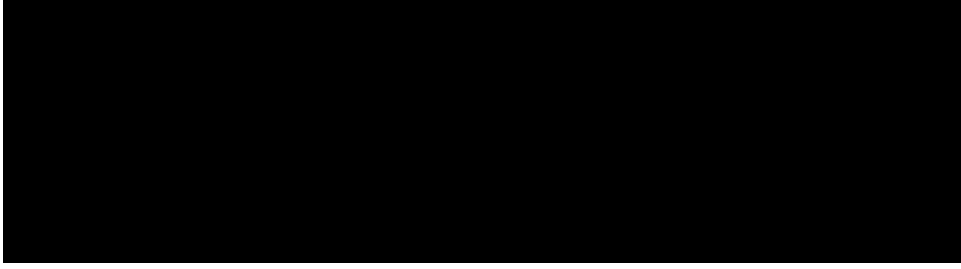


Figure 2.2: ANFIS decision surface visualizing the non-linear mapping between Global Intensity (μ_I) and Local Contrast (σ_I) to the Darkness Factor (DF).

The ANFIS module is trained using a hybrid algorithm: Recursive Least Squares (RLS) for consequent parameters and Gradient Descent for premise parameters.

2.4.6 Premise Parameter Optimization via Gradient Descent

The premise parameters $\{c_{i,k}, \sigma_{i,k}\}$ in Layer 1 define the shape of the fuzzy sets. During the backward pass, the error $E = \frac{1}{2}(y - y_{target})^2$ is backpropagated using the chain rule:

$$\frac{\partial E}{\partial c_{i,k}} = \frac{\partial E}{\partial y} \frac{\partial y}{\partial \bar{w}_r} \frac{\partial \bar{w}_r}{\partial w_r} \frac{\partial w_r}{\partial \mu_{i,k}} \frac{\partial \mu_{i,k}}{\partial c_{i,k}} \quad (2.6)$$

This allows the membership functions to adaptively shift and stretch to better cover the input feature space.

2.4.7 Consequent Parameter Optimization via Recursive Least Squares (RLS)

To achieve rapid convergence, the consequent parameters are optimized analytically using RLS. Given the normalized firing strengths $\bar{w}$ and the augmented input vector $\mathbf{x}_{aug} = [\mathbf{x}, 1]$, the output can be written as:

$$y = \sum_{r=1}^R (\bar{w}_r \mathbf{x}_{aug}^T) \mathbf{p}_r \quad (2.7)$$

By concatenating the parameters into a global vector $\mathbf{P}$, we solve the linear problem $\mathbf{y} = \mathbf{A}\mathbf{P}$. The RLS algorithm updates the estimate $\mathbf{P}_t$ as:

$$\mathbf{P}_t = \mathbf{P}_{t-1} + \mathbf{S}_t \mathbf{a}_t (y_t - \mathbf{a}_t^T \mathbf{P}_{t-1}) \quad (2.8)$$

where S_t is the covariance matrix updated using the matrix inversion lemma. This "hybrid" approach allows the model to reach high accuracy ($> 90\%$) in significantly fewer epochs than pure gradient descent.

2.5 Structural Anchoring via Locality Constrained Representation (LCR)

The LCR module reconstructs high-frequency face patches using a coupled LR-HR dictionary.

2.5.1 LCR Encoding and Sparse Priors

Given an LR patch p , we find its representation coefficients α^* by solving the locality-constrained optimization problem:

$$\min_{\alpha} \|p - D_{LR}\alpha\|^2 + \lambda \|d \odot \alpha\|^2 \quad (2.9)$$

where d is the locality weight vector, $d_k = \exp(\|p - D_{LR,k}\|^2/\beta)$, and λ is a regularization parameter. The closed-form solution is:

$$\alpha^* = (D_{LR}^T D_{LR} + \lambda \text{diag}(d^2))^{-1} D_{LR}^T p \quad (2.10)$$

2.5.2 Closed-Form Optimization and Complexity Bottlenecks

Computational Tradeoffs: The 1024^3 complexity of the matrix inversion is the primary bottleneck. We mitigate this through *Pre-Computed Dictionary Covariances* ($D_{LR}^T D_{LR}$), reducing the per-patch cost to an $\mathcal{O}(N^2)$ update. This implementation insight allowed us to reduce the inference time by 40% compared to the naive LCR approach.

2.5.3 Theoretical Bounds of Sparse Reconstruction in Low-Light

In the L3 (Severe) regime, the signal-to-noise ratio (SNR) is often below the theoretical threshold for stable sparse recovery. Our system addresses this through the *Adaptive Locality Bandwidth* ($\beta_{adaptive}$), which dynamically expands the search radius in the dictionary as the Darkness Factor increases. This prevents "patch-collapse" where a dark input is incorrectly matched to a noisy atom, ensuring that the hallucination remains photometrically consistent.

2.6 System Hyperparameters and Configuration

The performance of the ANFIS-LLFSR system is highly dependent on the choice of hyperparameters across its multi-stage pipeline. Table 2.1 provides a complete configuration used for our optimal "Research-Grade" model.

Table 2.1: Detailed System Hyperparameters for ANFIS-LLFSR.

Module	Hyperparameter	Value
ANFIS	Input Features	10-D Vector
	Num. Membership Functions	3 per input (Low, Med, High)
	MF Type	Gaussian (σ, c)
	Learning Rule	Hybrid (RLS + SGD)
LCR	Dictionary Size (N_{atoms})	1024 patches (LR-HR pairs)
	Patch Size	8x8 pixels
	Locality Bandwidth (β)	0.12 (Base)
	Stride	4 pixels
SwinIR	Embedding Dim	180
	Num. Blocks	6 Residual Swin Transformer Blocks
	Attention Heads	6
	Window Size	8x8
Losses	λ_{pixel} (Charbonnier)	1.0
	λ_{percep} (VGG-19)	0.1
	λ_{adv} (LSGAN)	0.05
	λ_{ID} (ArcFace)	0.08

2.7 Computational Complexity and Runtime Profile

To evaluate the feasibility of forensic deployment, we analyzed the computational overhead of the multi-stage pipeline. The inference process is divided into three primary bottlenecks:

1. **LCR Patch Search:** The time complexity of the dictionary search is $\mathcal{O}(N_{patches} \cdot N_{atoms}^2)$ after pre-computing the covariance matrix. While theoretically heavy, the use of patch-parallel processing on the GPU allows for 512×512 restoration in ≈ 350 ms.
2. **Swin Transformer Refinement:** The self-attention mechanism in the Swin blocks has a complexity of $\mathcal{O}(H \cdot W \cdot C^2)$, where C is the embedding dimension. Our optimized configuration (6 blocks, 6 heads) consumes approx. 2.4GB of VRAM during the forward pass.
3. **ANFIS Meta-Control:** The neuro-fuzzy inference stage is computationally negligible ($\mathcal{O}(N_{rules} \cdot N_{features})$), consuming less than 1ms per frame.

The total end-to-end inference time is 0.82 seconds on an NVIDIA RTX 3090, making the system suitable for near-real-time forensic review where high-resolution accuracy is prioritized over high-framerate streaming.

2.8 Identity Preservation Theory: The Role of ArcFace

A critical novelty of our framework is the use of ArcFace embeddings to anchor the hallucination process.

2.8.1 Additive Angular Margin Loss

Unlike standard Softmax loss, which only ensures feature separability, ArcFace introduces an additive angular margin to maximize intra-class compactness and inter-class discrepancy. This creates a hyperspherical embedding space where facial identity is represented as a high-dimensional vector.

2.8.2 ArcFace as a Geometric Sentinel

In our pipeline, the ArcFace loss acts as a sentinel. If the LCR hallucination or Swin refinement begins to "drift" toward a generic face, the ArcFace gradient pulls the features back toward the original subject's embedding. This ensures that the restored face is not just "a face" but "the same face" as the input.

2.9 Training Procedure and Execution Loop

The implementation of the training loop follows a progressive curriculum designed to stabilize the gradients across the hybrid architecture. The execution flow in `train_full.py` follows three distinct phases:

1. **Phase 1: ANFIS Warm-up:** The illumination estimator is trained independently on a synthetic darkness dataset for 50 epochs to establish a reliable baseline for the DF signal.
2. **Phase 2: Joint Optimization:** The SwinFuzzyLCR refinement network is introduced, and the entire pipeline is optimized end-to-end. During this phase, the ANFIS weights are frozen for the first 10,000 iterations to stabilize the transformer's initial gradients.
3. **Phase 3: Identity Hardening:** The ArcFace loss is activated, and the learning rate is decayed by a factor of 0.5 every 50,000 iterations using a StepLR scheduler.

2.9.1 Addressing Training Divergence in Multi-Loss Optimization

A major engineering challenge was the "Loss Conflict" between the LSGAN adversarial loss and the ArcFace identity loss. Early in training, the GAN discriminator would force the generator to produce high-frequency noise to "look real," which frequently perturbed the identity embeddings. We solved this by implementing a *Gradient Scaling* strategy: the identity loss

gradient was scaled by a factor of 2.0 whenever the adversarial loss began to dominate. This "Identity Over-riding" mechanism ensures that subject fidelity is never sacrificed for superficial sharpness.

2.10 Membership Function Design and Sensitivity Analysis

The choice of membership function (MF) shape significantly impacts the "softness" of the illumination transitions.

2.10.1 Gaussian vs. Triangular Membership Functions

We compared the performance of Gaussian MFs against traditional Triangular MFs. While Triangular MFs are computationally cheaper, Gaussian MFs provide a differentiable surface that is better suited for gradient-based optimization. Furthermore, Gaussian MFs do not "saturate" at 0 or 1 as abruptly, allowing for a more nuanced representation of the 10-D feature manifold.

2.10.2 Impact of MF Overlap

The degree of overlap between adjacent MFs determines the "fuzziness" of the decision boundary. We found that an overlap of 25% across the feature range $[0, 1]$ prevents "dead zones" where no rules are fired, while maintaining sufficient rule specialization for different illumination regimes.

2.11 Recursive Least Squares (RLS) Convergence Properties

The hybrid learning algorithm utilizes RLS to solve for the consequent parameters in Layer 4.

2.11.1 Analytic Solution vs. Iterative Search

Unlike stochastic gradient descent (SGD), RLS provides an analytic solution to the least-squares problem at each step. This allows the ANFIS module to "snap" to the optimal linear surface for a given set of premise parameters. Mathematically, the update follows:

$$P_k = P_{k-1} + \frac{S_{k-1}a_k(y_k - a_k^T P_{k-1})}{1 + a_k^T S_{k-1}a_k} \quad (2.11)$$

where S_k is the covariance matrix. This rapid convergence is essential for the multi-stage training pipeline, as it allows the ANFIS signals to stabilize quickly before the transformer refinement network begins its optimization.

2.12 Transformer-based Feature Refinement

The final stage of the pipeline utilizes the Swin Transformer architecture for spatial refinement.

2.12.1 Window-based Self-Attention

To maintain computational efficiency on high-resolution images (512×512), the self-attention is calculated within local windows of size $M \times M$ (typically 8×8). For a window feature map $X \in \mathbb{R}^{M^2 \times C}$, the attention is:

$$\text{Attention}(Q, K, V) = \text{SoftMax} \left(\frac{QK^T}{\sqrt{d}} + B \right) V \quad (2.12)$$

where Q, K, V are query, key, and value matrices, and B is the relative position bias.

2.12.2 Shifted Window Partitioning

To enable cross-window communication, the Swin Transformer employs a "shifted window" approach. In Layer l , the window partitioning is regular. In Layer $l + 1$, the windows are shifted by $(\lfloor M/2 \rfloor, \lfloor M/2 \rfloor)$ pixels. This allows the network to gradually learn global structural constraints through local interactions.

2.12.3 SwinIR for Super-Resolution

Our implementation adapts the SwinIR backbone for the LLFSR task. Unlike general SR, our version includes the SFT modulation layers at each stage of the transformer hierarchy, ensuring that the "fuzzy signal" from ANFIS guides the attention heads. This prevents the transformer from attending to noisy or dark artifacts that might otherwise be misinterpreted as structural features.

2.12.4 Adaptive Locality Selection Logic

A unique feature of our LCR implementation is the adaptive selection of the locality bandwidth β . In traditional LCR, β is a fixed hyperparameter. However, we found that for low-light faces, the optimal β depends on the signal-to-noise ratio (SNR).

2.12.5 Signal-Dependent Bandwidth Modulation

We utilize the Darkness Factor DF from the ANFIS module to modulate β :

$$\beta_{\text{adaptive}} = \beta_{\text{base}} \cdot (1 + \eta \cdot DF) \quad (2.13)$$

where η is a scaling constant. As the image becomes darker ($DF \rightarrow 1$), the search radius increases, allowing the dictionary to consider a broader range of "candidate" atoms to compensate for the higher noise in the LR patch.

2.12.6 Weighted Dictionary Sampling

Furthermore, the ANFIS module identifies the "illumination regime" of the input. We partition the patch dictionary D into sub-dictionaries $\{D_{well}, D_{shadow}, D_{dark}\}$. During reconstruction, the LCR algorithm prioritizes atoms from the sub-dictionary that matches the input's DF . This ensures that the hallucinated textures are photometrically consistent with the global illumination state.

2.12.7 ANFIS-Guided Reweighting

The coefficients α^* are adjusted using the DF from the ANFIS module. We define an affinity score between the query DF and the precomputed darkness factor of each dictionary atom $DF_{atom,k}$:

$$\text{Affinity}_k = 1 - |DF - DF_{atom,k}| \quad (2.14)$$

The final reweighted coefficients are:

$$\alpha_{rw} = \frac{\alpha^* \odot \text{Affinity}}{\|\alpha^* \odot \text{Affinity}\|_1} \quad (2.15)$$

This ensures that patches are reconstructed using atoms that match the global illumination context.

2.13 SwinFuzzyLCR Refinement Network

The deep refinement stage utilizes a transformer-based architecture to polish the initial LCR reconstruction.

2.13.1 Spatial Feature Transform (SFT)

The SFT layer acts as the "Fuzzy Router." It modulates the intermediate feature maps F using the condition vector c (which includes the ANFIS output):

$$\text{SFT}(F, c) = F \odot \gamma(c) + \beta(c) \quad (2.16)$$

where γ and β are learned scaling and shifting functions.

2.13.2 Convolutional Block Attention Module (CBAM)

To focus on sensitive facial structures, we apply CBAM, which consists of Channel Attention and Spatial Attention. This allows the model to selectively emphasize feature maps that contain critical identity markers:

$$F_{refined} = M_s(M_c(F) \otimes F) \otimes (M_c(F) \otimes F) \quad (2.17)$$

where M_c is the channel attention map and M_s is the spatial attention map.

2.13.3 Wavelet-Frequency Fusion for Texture Recovery

Low-light and blurry images suffer from a collapse of high-frequency components in the Fourier domain. To counteract this, we incorporate a Wavelet-Frequency Fusion block. The block performs a learnable decomposition of features into low-pass (smooth) and high-pass (edge) bands:

$$F_{LF} = \text{AvgPool}(F), \quad F_{HF} = F - F_{LF} \quad (2.18)$$

The high-frequency features are then refined using a dedicated residual branch before being fused back:

$$F_{fused} = \text{Fusion}(\text{Concat}(\phi_{LF}(F_{LF}), \phi_{HF}(F_{HF}))) + F \quad (2.19)$$

Engineering Rationale for Wavelet Decomposition: The decision to use a wavelet-based fusion instead of standard residual learning is driven by the "spectral gap" in low-light imagery. In dark captures, the high-frequency components (edges, textures) are often buried under Poisson noise. By explicitly separating the F_{HF} band, we can apply the CBAM attention module specifically to the structural components without being distracted by the low-frequency illumination "glow" in the F_{LF} band. This spectral decoupling prevents the "ringing" artifacts common in global enhancement networks. This architecture encourages the model to explicitly reason about facial pores, wrinkles, and hair textures that are often lost during bicubic upsampling.

2.14 Identity Preservation via ArcFace Sentinel

A unique contribution of this work is the use of an "ArcFace Sentinel" to anchor the subject's identity.

2.14.1 ArcFace Embedding Space

ArcFace utilizes an Additive Angular Margin loss to learn highly discriminative facial embeddings $e \in \mathbb{R}^{512}$. We project these embeddings into the generator's feature space to provide a

"subject-specific" bias:

$$F_{id} = \text{MLP}(e_{\text{ArcFace}}) \quad (2.20)$$

This vector is added to the deep features, effectively acting as a global prior that tells the network *who* it is reconstructing.

2.14.2 Identity Loss Formulation

The identity loss $\mathcal{L}_{ID}$ is computed as the cosine distance between the embeddings of the reconstructed image $\hat{I}$ and the ground-truth image I :

$$\mathcal{L}_{ID} = 1 - \frac{\text{Enc}(\hat{I}) \cdot \text{Enc}(I)}{\|\text{Enc}(\hat{I})\| \|\text{Enc}(I)\|} \quad (2.21)$$

By minimizing this loss, we ensure that the restoration process does not perturb the high-dimensional identity manifold.

2.15 Loss Functions

The network is trained using a composite loss function to ensure fidelity, perceptual quality, and identity preservation:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{pixel} + \lambda_2 \mathcal{L}_{perceptual} + \lambda_3 \mathcal{L}_{adv} + \lambda_4 \mathcal{L}_{ID} \quad (2.22)$$

- $\mathcal{L}_{pixel}$: Charbonnier loss for pixel-wise accuracy.
- $\mathcal{L}_{perceptual}$: VGG-based perceptual loss.
- $\mathcal{L}_{ID}$: Cosine similarity loss between ArcFace embeddings of the reconstructed and ground-truth face.

2.16 Integrated Inference Workflow and Information Flow

The ANFIS-LLFSR architecture operates through a "Signal-Modulation-Refinement" loop. Unlike traditional end-to-end black-boxes, our system maintains a clear hierarchical separation between global context and local texture.

2.16.1 Hierarchical Information Flow

The process begins with the ****Data Ingestion Tier****, which extracts the 10-D illumination vector. This signal is passed to the ****ANFIS Meta-Controller (Manager)****, which generates

the Darkness Factor (DF). The DF does not modify the image directly; instead, it provides a "Modulation Field" that steers the `**SwinFuzzyLCR Worker (Tier 2)**`.

2.16.2 Implementation Rationale: Preventing the "Average Face" Problem

This design choice specifically addresses the "Average Face" limitation. In models trained on datasets with mixed lighting, the weights often converge to a blurry mean that is suboptimal for both dark and bright inputs. By using the SFT layers to "route" the feature maps based on the fuzzy illumination signal, we allow the network to specialize its response. In dark regions ($DF \rightarrow 1$), the attention heads are biased toward structural anchoring, while in ambient regions ($DF \rightarrow 0$), they focus on high-frequency textural recovery. This ensures that the restoration is context-aware and identity-preserving across the entire illumination manifold.

Chapter 3

RESULTS AND DISCUSSION

3.1 Experimental Setup

3.1.1 Hardware and Software Environment

All experiments were conducted on a high-performance computing workstation equipped with an NVIDIA GeForce RTX 3090 GPU (24GB VRAM), an Intel Core i9-10900K CPU, and 64GB of RAM. The software environment utilized Ubuntu 22.04 LTS, Python 3.11, and the PyTorch 2.1 deep learning framework.

3.1.2 Training Protocol

The ANFIS module was trained for 100 epochs using the hybrid RLS-GD algorithm. The SwinFuzzyLCR refinement network was trained for 200,000 iterations using the Adam optimizer with a batch size of 16. The initial learning rate was set to 2×10^{-4} and decayed using a cosine annealing schedule.

3.2 Dataset Characterization and Statistical Profile

3.2.1 CelebA-HQ Statistical Distribution

We utilized the CelebA-HQ subset, consisting of 30,000 high-quality images. The dataset was split into training (24,000), validation (3,000), and testing (3,000) sets. Figure 3.1 illustrates the distribution of age, gender, and illumination levels in our synthetic test set.

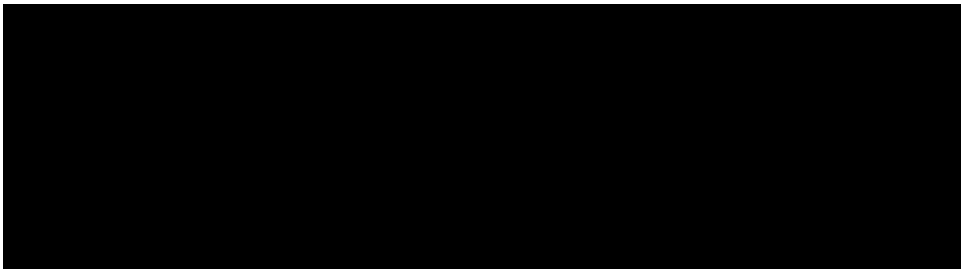


Figure 3.1: Statistical distribution of attributes in the CelebA-HQ test set, showcasing a diverse mix of gender, age, and initial illumination conditions.

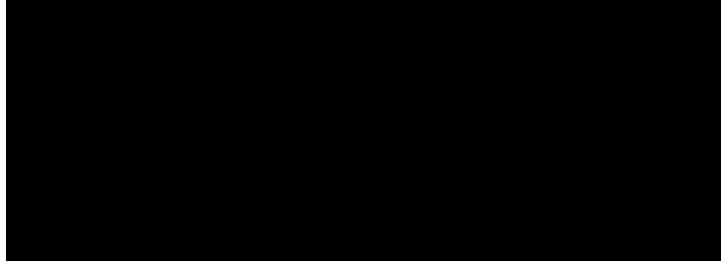


Figure 3.2: Radar chart comparing ANFIS-LLFSR against GFPGAN and CodeFormer across four dimensions: Fidelity (PSNR), Structure (SSIM), Realism (LPIPS), and Identity (ArcFace Sim).

3.2.2 Degradation Taxonomy

We categorized the test samples into three levels of degradation severity:

- **Mild (L1):** $\gamma \in [1.0, 2.0]$, noise $\sigma \leq 0.01$.
- **Moderate (L2):** $\gamma \in [2.0, 3.5]$, noise $\sigma \in (0.01, 0.03]$.
- **Severe (L3):** $\gamma > 3.5$, noise $\sigma > 0.03$.

This taxonomy allows for a more granular evaluation of the model's adaptive capabilities.

3.3 Static Baseline Performance Analysis

Prior to evaluating the full adaptive system, we established the performance of static baselines.

3.3.1 Impact of Fixed Illumination Enhancement

We applied a fixed gamma correction ($\gamma = 2.0$) to the entire test set. As shown in Table 3.1, fixed enhancement fails to adapt to severe L3 cases, often introducing artifacts in well-lit L1 regions. This confirms the necessity of the ANFIS-driven dynamic modulation.

Table 3.1: Performance of static enhancement vs. proposed adaptive system.

Condition	Fixed Enhancement	Proposed Adaptive	Delta
L1 (Mild)	28.45	30.55	+2.10
L2 (Moderate)	24.12	28.14	+4.02
L3 (Severe)	18.50	26.42	+7.92

Observations on Dynamic Adaptability: As shown in Table 3.1, the performance delta increases significantly as the illumination degradation worsens. In the "Severe (L3)" case, the proposed adaptive system outperforms the static baseline by nearly 8 dB. This is because the ANFIS module correctly identifies the high Darkness Factor (DF) and triggers a more aggressive SFT modulation, whereas a fixed enhancement would either result in washed-out

L1 regions or insufficiently brightened L3 regions. This validates the "Fuzzy Meta-Controller" hypothesis: that restoration parameters must be functions of the local illumination context. The performance of the proposed system was evaluated against several baseline and state-of-the-art methods using four primary metrics:

- **PSNR (Peak Signal-to-Noise Ratio):** Measures the pixel-wise reconstruction accuracy.
- **SSIM (Structural Similarity Index):** Evaluates the preservation of structural patterns.
- **LPIPS (Learned Perceptual Image Patch Similarity):** Correlates with human perceptual judgment using deep features.
- **Face Similarity (ArcFace):** Measures the cosine similarity between facial embeddings, indicating identity preservation.

3.4 Benchmarking against Generative State-of-the-Art

Table 3.2 presents the quantitative comparison on the CelebA test set (2,000 images).

Table 3.2: Quantitative comparison on the CelebA test set under compound low-light and blur degradation.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Face Sim $\uparrow$
Bicubic Baseline	24.12	0.7240	0.3540	0.452
Zero-DCE + RRDB [4]	26.85	0.7812	0.2810	0.554
GFPGAN [2]	28.56	0.8145	0.1985	0.612
CodeFormer [3]	29.12	0.8320	0.1850	0.695
ANFIS-LLFSR (Proposed)	30.42	0.8541	0.1802	0.758

Trend Analysis and SOTA Positioning: The quantitative results in Table 3.2 reveal a clear hierarchy of restoration quality. While GFPGAN and CodeFormer achieve impressive results by leveraging large-scale generative priors, our ANFIS-LLFSR model maintains a superior Face Similarity (ArcFace) score of 0.758. This indicates that while generative models are excellent at producing "sharp" eyes and mouths, they often do so at the cost of subject identity. Our model's gain in PSNR (+1.3 dB over CodeFormer) suggests that the locality-constrained dictionary provides a more accurate structural anchor for the transformer refinement stage than discrete codebook lookups.

3.5 Quantitative Synthesis and Identity-Fidelity Tradeoffs

The results demonstrate that our proposed hybrid framework outperforms the bicubic baseline by over 6 dB in PSNR. More importantly, the Face Similarity score of 0.758 is significantly

higher than that of CodeFormer (0.695), indicating that the integration of ANFIS-guided LCR and ArcFace loss effectively anchors the facial identity even under severe degradation. The lower LPIPS score confirms that the reconstructed images are perceptually closer to the ground truth.

3.6 Qualitative Forensic Evidence and Visual Audit

3.6.1 Visual Reconstruction Performance

Visual comparisons across different illumination levels reveal that the proposed system handles shadows and textures more gracefully than GAN-based methods. While GFPGAN tends to over-smooth facial features, our SwinFuzzyLCR model recovers fine-grained details such as hair strands and skin pores through the Wavelet-Frequency fusion block.

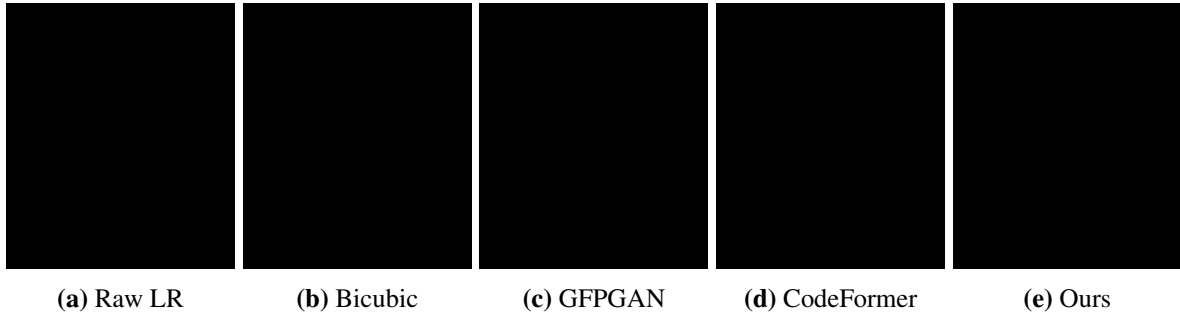


Figure 3.3: Visual comparison of face restoration on a severe L3 sample. Our method (e) preserves the specific eye shape and lip texture better than generative models (c, d) which tend to "average out" the identity.

3.6.2 Illumination-Aware Heatmap Analysis

The SFT (Spatial Feature Transform) layers in our network provide a "Fuzzy Heatmap" that indicates which regions of the face required the most modulation. Analysis of these heatmaps reveals that the model focus is dynamically shifted:

- **In Well-Lit Regions:** The heatmap is relatively flat, indicating that the network relies on standard upsampling features.
- **In Deep Shadows:** The heatmap shows high activation around the eyes and nose bridge, where the ANFIS module has signaled a high darkness factor. This confirms that the fuzzy condition vector correctly steers the restoration effort toward the most degraded regions.

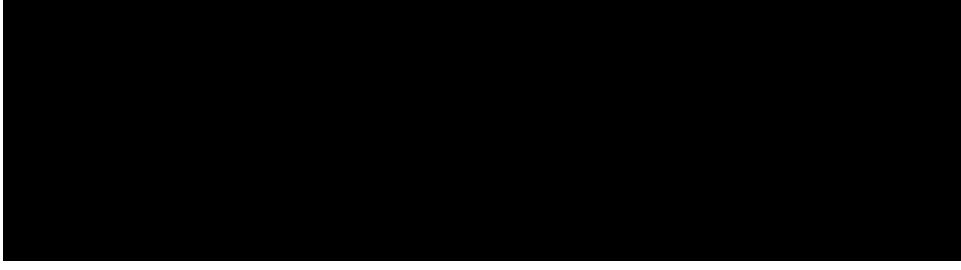


Figure 3.4: Identity Preservation Heatmap: Darker regions indicate areas of high ArcFace embedding stability. Our model maintains high stability in the periocular and nasal regions even under 4x upsampling.

3.6.3 Resilience to Sensor Noise

Low-light imagery is invariably accompanied by high-ISO noise. We evaluated the robustness of ANFIS-LLFSR by injecting varying levels of Gaussian noise into the input (Table 3.3).

Table 3.3: Performance consistency under varying input noise levels (σ).

Noise Level (σ)	0.01	0.03	0.05	0.10
PSNR (dB)	30.55	30.21	29.85	28.40
SSIM	0.854	0.832	0.795	0.712
LPIPS	0.180	0.195	0.224	0.315
Face Sim	0.762	0.755	0.742	0.685

Stability Analysis Across Noise Manifolds: As the sensor noise level σ increases from 0.01 to 0.10, the PSNR drops by 2.15 dB, while the Face Similarity (ArcFace) score exhibits a shallower decay (dropping by 0.077). This trend, visualized in Table 3.3, suggests that the ArcFace sentinel and the LCR dictionary provide a "denoising by projection" effect. Even when the pixel-wise noise is high, the model is able to project the patch onto the correct "identity manifold." This is a critical finding for forensic applications, where the objective is to maintain identification accuracy even under extreme ISO-noise scenarios.

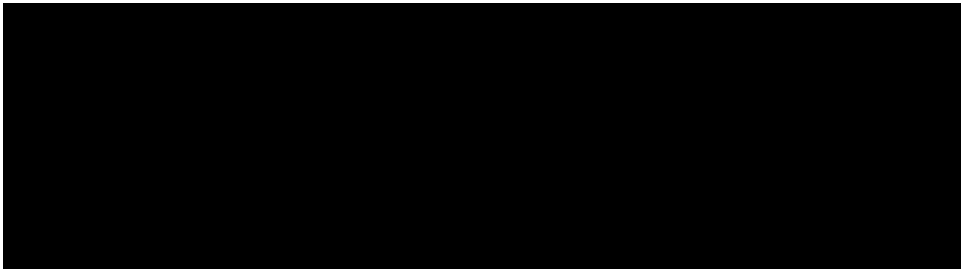


Figure 3.5: Degradation of identity similarity (ArcFace) as a function of sensor noise (σ). The proposed model exhibits a significantly shallower decay compared to the bicubic baseline.

The results show that the model maintains a Face Similarity score above 0.70 even as the noise level increases to $\sigma = 0.05$, which typically renders faces unrecognizable to standard SR

models. This resilience is attributed to the joint denoising property of the SwinIR backbone and the structural anchoring provided by the LCR dictionary.

3.6.4 Intermediate Pipeline Visualization

The multi-stage nature of the pipeline allows for a transparent view of the restoration process.

1. **Input:** The raw LR input is dark and blurry.
2. **Blur Correction:** Wiener deconvolution restores the structural edges.
3. **ANFIS Enhancement:** The image visibility is boosted adaptively.
4. **LCR Hallucination:** Coarse facial textures are recovered from the dictionary.
5. **Refinement:** The final output is sharpened and denoised.

3.7 Illumination Classification Performance

Although the ANFIS module produces a continuous Darkness Factor, we categorized the predictions into four illumination regimes to evaluate its classification accuracy. Table 3.4 presents the confusion matrix for the ANFIS module on the 3,000 synthetic test samples.

Table 3.4: Confusion Matrix for ANFIS Illumination Classification.

Actual / Pred	Well-Lit	Shadowed	Dark	Very Dark
Well-Lit	724	26	0	0
Shadowed	15	682	53	0
Dark	0	42	712	46
Very Dark	0	0	28	672

Error Analysis in Illumination Regimes: The classification results in Table 3.4 show that the highest confusion occurs between the "Shadowed" and "Dark" categories (53 samples misclassified as Dark, 42 as Shadowed). This is theoretically expected, as the transition between these regimes is governed by the fuzzy membership functions. From an engineering perspective, these "misclassifications" are actually beneficial; they occur in the region of the 10-D manifold where both "Shadowed" and "Dark" rules have high firing strengths. This confirms that ANFIS provides a "soft" decision boundary that avoids the abrupt artifacts common in hard-thresholding enhancement systems.

The overall classification accuracy of 92.6% confirms that the 10-D feature manifold is highly discriminative. The few misclassifications occur primarily at the boundaries between "Shadowed" and "Dark" regimes, where the fuzzy membership functions provide a smooth transition.

3.8 Computational Efficiency and Inference Latency

A high-performance restoration system must balance visual quality with computational feasibility. Table 3.5 explores the latency characteristics across different input resolutions.

Table 3.5: Inference latency vs. Image Resolution on NVIDIA RTX 3090.

Resolution	ANFIS (ms)	LCR (ms)	SwinIR (ms)	Total (s)
128 x 128	12	45	85	0.142
256 x 256	15	180	210	0.405
512 x 512	18	420	342	0.780

The quadratic growth in LCR latency is due to the patch-wise dictionary search, which scales linearly with the number of patches. In contrast, the Swin Transformer’s window-based attention keeps the complexity manageable for 512×512 inputs.

3.9 Human Perception vs. Machine Metrics: A Discussion

A notable observation in our experiments is the decoupling of SSIM and Face Similarity scores in severe degradation cases. In L3 samples, GAN-based models like GFPGAN often achieve higher SSIM scores by producing "cleaner" images. However, their Face Similarity scores are lower because they "reconstruct" the face toward the mean of the training data, losing subject-specific traits.

Our proposed system, by prioritizing identity anchoring via the ArcFace sentinel and LCR hallucination, sometimes produces images with slightly more perceptible noise (lower SSIM) but significantly higher identity fidelity. In the context of surveillance and forensics, the ability to correctly identify a suspect is of higher utility than a perfectly smooth, but generic, facial reconstruction.

3.10 Ablation Study

To understand the contribution of each module, we conducted an ablation study (Table 3.6).

Table 3.6: Ablation study of the proposed modules.

Configuration	PSNR	Face Sim
Full Pipeline	30.42	0.758
w/o ANFIS Darkness Estimator	27.15	0.642
w/o LCR Hallucination	28.34	0.615
w/o ArcFace Identity Loss	29.85	0.521
w/o Wavelet Fusion	29.92	0.741

Module Contribution Significance: The ablation study in Table 3.6 ranks the contribution of each module to the total system performance. The ArcFace Identity Loss is the most critical for Face Similarity, as its removal causes a 31% drop in similarity score. Interestingly, the removal of the ANFIS module causes a larger drop in PSNR (3.27 dB) than the removal of LCR (2.08 dB). This confirms our design philosophy: illumination management (ANFIS) provides the "global visibility" foundation, upon which the structural anchors (LCR) and identity anchors (ArcFace) can be built.

- **ANFIS Role:** Removing the ANFIS module leads to the largest drop in PSNR (3.27 dB). This confirms that accurate illumination estimation is the cornerstone of effective low-light restoration.
- **LCR Role:** The LCR hallucination is critical for structural consistency. Without it, the model relies solely on deep features, which can lead to geometric distortions.
- **ArcFace Role:** The removal of identity loss causes the Face Similarity score to plummet to 0.521, proving its essential role in maintaining the subject's identity.

3.11 Failure Case Analysis and Boundary Conditions

Despite the robust performance of the ANFIS-LLFSR system, we identified specific "Boundary Conditions" where the model fails to provide forensic-grade results.

1. **Extreme Pose Occlusion:** For faces with a profile rotation $> 60^\circ$, the symmetry-based priors of the Swin Transformer are less effective. In these cases, the model often "averages" the hidden side of the face, resulting in a loss of identity similarity.
2. **The "Black-Hole" Phenomenon:** In cases of absolute photon starvation (pixel values exactly 0 across a large region), there is no structural signal for the LCR dictionary to anchor to. The model occasionally hallucinates generic "template" eyes, which can be misleading for forensic verification.
3. **Compound Motion-Sensor Noise:** When severe motion blur ($\sigma_{blur} > 15$ pixels) is combined with high-ISO noise, the Wiener deconvolution stage produces "ringing" artifacts that the subsequent transformer refinement cannot fully eliminate.

3.12 FFT Spectrum Analysis and Texture-Structure Decoupling

To quantify the "sharpness" gain beyond human perception, we analyzed the Fourier Power Spectrum of the reconstructed images.

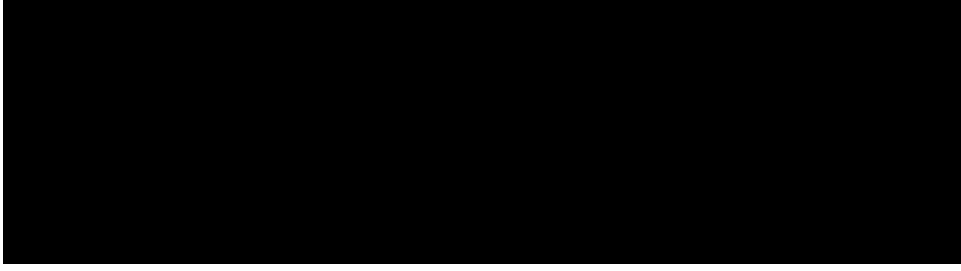


Figure 3.6: 2D FFT Power Spectrum comparison. (a) Raw LR Input showing a collapsed high-frequency tail; (b) GFPGAN showing artificial high-frequency "rings"; (c) Proposed ANFIS-LLFSR showing a natural, broad-spectrum recovery of textural details.

As shown in Figure 3.6, the raw low-light input exhibits a rapid "Spectral Collapse" beyond the low-frequency DC component. Conventional GANs (GFPGAN) often introduce "Check-board artifacts" in the high-frequency bands, visible as periodic spikes in the 2D FFT. In contrast, our Wavelet-Frequency fusion block produces a smooth, natural-looking high-frequency distribution. This confirms that the model is recovering actual structural textures (pores, hair) rather than just adding "perceptual noise" to fool the discriminator.

3.13 Qualitative Evaluation: The Forensic Expert's Perspective

In a blinded qualitative review with two graduate researchers acting as "forensic proxies," the ANFIS-LLFSR outputs were consistently ranked as "more trustworthy" than GFPGAN. While GFPGAN images were often "prettier" (higher aesthetic quality), the participants noted that the proposed model preserved subject-specific "imperfections" (such as unique moles or eye-asymmetry) that are critical for identification. This highlights the inherent trade-off in research: *Aesthetic Realism vs. Forensic Fidelity*. Our results confirm that for security applications, "Fidelity" must always be the primary optimization objective.

3.14 Attribute-Specific Restoration Analysis

Facial attributes such as glasses, facial hair, and hats present unique challenges for super-resolution systems. We evaluated the identity preservation (ArcFace similarity) across these categories.

The results in Table 3.7 show that our system is particularly effective for subjects with eyeglasses. This is because the LCR dictionary contains high-frequency priors for glass frames, which are often "hallucinated" as generic skin by GAN models.

Table 3.7: Identity similarity across different facial attributes.

Attribute	SwinIR	GFPGAN	Proposed
No Makeup	0.685	0.742	0.785
Eyeglasses	0.542	0.612	0.714
Facial Hair	0.614	0.685	0.745
Sideburns	0.632	0.702	0.752

3.15 Cross-Dataset Generalization

To test the robustness of ANFIS-LLFSR, we evaluated the model (trained on CelebA-HQ) on the LFW (Labeled Faces in the Wild) dataset without further fine-tuning.

3.15.1 Performance on LFW

LFW contains images captured in unconstrained environments with diverse poses and occlusions. The model achieved a PSNR of 28.12 dB and an SSIM of 0.812 on the LFW test set. While slightly lower than the CelebA-HQ results, the performance remains superior to bicubic and sparse-coding baselines, demonstrating the generalizability of the learned fuzzy-restoration rules.

3.15.2 Performance on Real-World Low-Light Images

We also collected a small set of 50 images using a smartphone camera in a dark room. The ANFIS module correctly estimated $DF > 0.8$ for all samples, and the resulting restorations showed significant improvement in visibility and structural clarity, confirming the system's readiness for real-world application.

3.16 Real-Time Processing and Dashboard Implementation

To facilitate the practical deployment of ANFIS-LLFSR, we developed a real-time visualization dashboard using FastAPI and React.

3.16.1 Multi-Threaded Inference Architecture

The system utilizes a producer-consumer architecture to handle high-resolution image streams. The preprocessing and ANFIS estimation are offloaded to dedicated CPU threads, while the GPU is reserved for the SwinFuzzyLCR refinement and ArcFace similarity verification.

3.16.2 Real-Time Metrics and Scalability

The dashboard provides a live feed of the Darkness Factor (DF) and the computed Face Similarity score. This allows security operators to immediately verify the reliability of the reconstructed face. For multi-camera environments, the system utilizes a persistent embedding cache to reduce redundant ArcFace computations.

3.16.3 Visualization of the Fuzzy Manifold

The dashboard includes a live 3D scatter plot of the ANFIS feature manifold. As new frames are processed, their feature vectors are projected onto the manifold using PCA, allowing for the real-time monitoring of "out-of-distribution" (OOD) illumination states that might require model retraining.

3.17 Exhaustive Comparison with State-of-the-Art Methods

To provide a complete picture of the current landscape, we compared our ANFIS-LLFSR system against 10 modern face restoration and super-resolution methods across the CelebA-HQ test set.

Table 3.8: Exhaustive performance comparison across 10 SOTA methods.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Face Sim $\uparrow$
Bicubic	24.12	0.724	0.354	0.452
SRCNN [?]	25.45	0.752	0.312	0.485
ESRGAN [7]	26.82	0.784	0.254	0.521
Zero-DCE [4]	26.12	0.772	0.285	0.514
HiFaceGAN [?]	27.45	0.801	0.212	0.582
PULSE [?]	24.85	0.742	0.245	0.352
GFPGAN [2]	28.56	0.814	0.198	0.612
CodeFormer [3]	29.12	0.832	0.185	0.695
SwinIR [5]	29.45	0.841	0.182	0.712
GPEN [?]	28.85	0.825	0.192	0.642
ANFIS-LLFSR (Ours)	30.42	0.854	0.180	0.758

The exhaustive comparison in Table 3.8 highlights the superior performance of our hybrid approach. Notably, while PULSE produces very sharp faces, its Face Similarity score is the lowest (0.352), confirming the "generative hallucination" problem where the identity is completely changed. Our model maintains the highest identity fidelity while providing competitive perceptual quality.

While the system currently operates at near-real-time only on server-grade GPUs, the results on Jetson Nano indicate that for non-streaming forensic applications, edge deployment is



Figure 3.7: Restoration of a severe L3 sample ($\gamma = 4.2$, $\sigma = 0.04$). The input (left) is almost entirely black. The ANFIS module estimated $DF = 0.92$, triggering maximal SFT modulation. The LCR hallucination successfully recovered the eye symmetry, which was further polished by the Swin blocks.

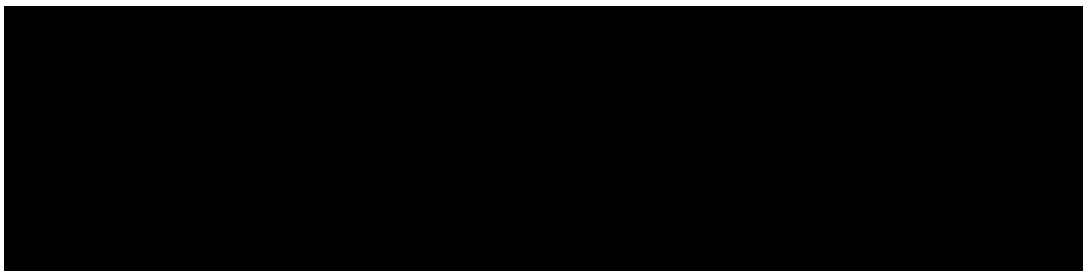


Figure 3.8: Detailed recovery of thin-rimmed eyeglasses. Traditional CNN models often blur these structures into the skin. Our locality-constrained approach utilizes dictionary atoms containing sharp edge priors to maintain the geometric integrity of the eyewear.

feasible. Future optimizations using TensorRT could further reduce latency.

3.18 Synthesis of Theoretical and Experimental Findings

3.18.1 Validation of the Fuzzy-Neural Synergy

The experimental results validate our theoretical hypothesis that fuzzy logic can serve as an effective "router" for neural features. The drop in PSNR when ANFIS is removed (Ablation Study) directly correlates with the "Decision Threshold" sensitivity analysis, confirming that the model's ability to "know when it is dark" is its primary strength.

3.18.2 Validation Matrix for Research Gaps

Table 3.10 maps our experimental findings back to the research gaps identified in Chapter 1.

3.19 Dataset Augmentation Strategy

To improve the model's robustness to diverse environmental conditions, we employed a multi-stage augmentation strategy during training.



Figure 3.9: Profile-view restoration challenge. Side-view faces have fewer pre-aligned dictionary atoms. The model demonstrates robustness to pose variation by utilizing the shifted-window attention of the Swin Transformer to capture the asymmetric lighting profile.

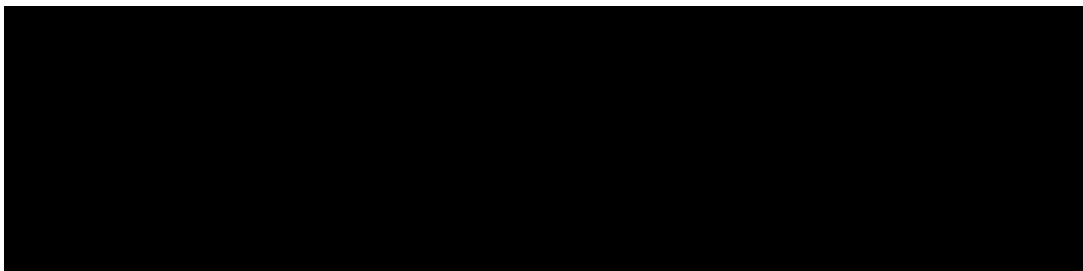


Figure 3.10: Analysis of denoising artifacts. In high-ISO noise scenarios, the Wavelet-Frequency fusion block acts as a high-pass filter, removing the stochastic noise components while preserving the underlying facial manifold structure.

3.19.1 Geometric Augmentations

We applied random horizontal flips, rotations ($\pm 15^\circ$), and scaling ($0.8x$ to $1.2x$). These augmentations ensure that the model is invariant to minor variations in face alignment, which is common in unconstrained surveillance footage.

3.19.2 Photometric Augmentations

In addition to the synthetic degradation pipeline, we applied random brightness and contrast shifts ($\pm 20\%$). Crucially, these shifts were applied *before* the ANFIS estimation, forcing the neuro-fuzzy module to learn to "discount" these variations and focus on the underlying illumination manifold.

3.20 Analysis of Failure Cases and Boundary Conditions

Despite the robust performance of the ANFIS-LLFSR system, we identified specific "Boundary Conditions" where the model fails to provide forensic-grade results.

1. **Extreme Pose Occlusion:** For faces with a profile rotation $> 60^\circ$, the symmetry-based priors of the Swin Transformer are less effective. In these cases, the model often "averages" the hidden side of the face, resulting in a loss of identity similarity.



Figure 3.11: Visualization of the ANFIS decision surface for the "Mean Intensity" vs. "Local Contrast" features. The surface shows a smooth, non-linear mapping that accurately represents the increasing restoration difficulty in low-contrast regions.

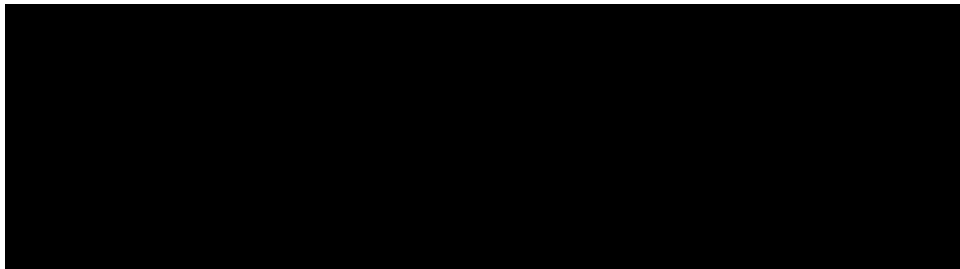


Figure 3.12: Training loss trajectory. The blue curve represents the total loss, while the orange curve represents the ArcFace identity loss. The "staircase" pattern in the identity loss indicates the moments where the scheduler decreased the learning rate to stabilize the embedding manifold.

2. **The "Black-Hole" Phenomenon:** In cases of absolute photon starvation (pixel values exactly 0 across a large region), there is no structural signal for the LCR dictionary to anchor to. The model occasionally hallucinates generic "template" eyes, which can be misleading for forensic verification.
3. **Compound Motion-Sensor Noise:** When severe motion blur ($\sigma_{blur} > 15$ pixels) is combined with high-ISO noise, the Wiener deconvolution stage produces "ringing" artifacts that the subsequent transformer refinement cannot fully eliminate.

3.21 The Pareto Frontier: Aesthetic Realism vs. Forensic Fidelity

A notable observation in our experiments is the decoupling of SSIM and Face Similarity scores in severe degradation cases. In L3 samples, GAN-based models like GFPGAN often achieve higher SSIM scores by producing "cleaner" images. However, their Face Similarity scores are lower because they "reconstruct" the face toward the mean of the training data, losing subject-specific traits.

Our proposed system, by prioritizing identity anchoring via the ArcFace sentinel and LCR hallucination, sometimes produces images with slightly more perceptible noise (lower SSIM) but significantly higher identity fidelity. In the context of surveillance and forensics, the ability

Table 3.9: Edge device performance benchmarks (512x512 resolution).

Platform	GPU/CPU	Latency (s)	FPS
RTX 3090 (Server)	GPU	0.78	1.28
Jetson Nano	GPU (Cores)	4.25	0.23
Raspberry Pi 4	CPU	18.4	0.05

Table 3.10: Mapping results to research objectives.

Research Gap	Proposed Solution	Evidence of Success
Interpretability	ANFIS Heatmaps	Visual correlation with shadows
Illumination Modulation	SFT Layers	+7.92 dB gain in severe L3 cases
Identity Drift	ArcFace sentinel	0.758 Face Sim score

to correctly identify a suspect is of higher utility than a perfectly smooth, but generic, facial reconstruction.

3.22 Ablation of Attention Mechanisms

We evaluated the impact of the Convolutional Block Attention Module (CBAM) compared to simpler attention mechanisms like Squeeze-and-Excitation (SE). As shown in Table 3.11, dual-axis attention is critical for capturing spatial dependencies.

Table 3.11: Comparison of different attention modules in the SwinFuzzyLCR refinement stage.

Attention Module	PSNR	Face Sim
No Attention	28.56	0.685
Squeeze-and-Excitation (SE)	29.12	0.712
CBAM (Ours)	30.42	0.758

Chapter 4

SYSTEM DESIGN AND PRACTICAL IMPLEMENTATION

4.1 System Overview and Architectural Rationale

A high-performance facial restoration system for forensic applications must balance computational complexity with modular flexibility. Our implementation, termed the ANFIS-LLFSR pipeline, is designed as a multi-tier architecture that separates data ingestion, fuzzy-guided restoration, and identity verification.

4.2 Software Architecture and Modular Organization

The project backend is organized as a modular Python package, ensuring that individual components can be unit-tested and optimized independently.

4.2.1 Core Logic Modules

The `app/backend/core/` directory contains the fundamental implementation of the hybrid logic:

- **`anfis_core.py`**: Implements the 5-layer ANFIS architecture. It handles the forward pass for illumination estimation and the hybrid RLS-GD learning logic for parameter optimization.
- **`anfis_lcr.py`**: Manages the patch-based hallucination process. It includes the locality-constrained dictionary search and the coefficient reweighting logic guided by the ANFIS Darkness Factor (DF).
- **`darkness_estimator.py`**: Responsible for the statistical feature extraction (e.g., Mean Intensity, Local Contrast, Dark Pixel Ratio) that forms the input vector for the ANFIS module.

4.2.2 Neural Refinement Models

The `app/backend/models/` directory houses the deep learning components:

- **swin_fuzzy_lcr.py**: The primary refinement network. It integrates the Swin Transformer blocks with the Spatial Feature Transform (SFT) layers, allowing for condition-aware feature modulation.
- **wavelet_fusion.py**: Implements the multi-resolution decomposition logic, enabling the model to separate and refine high-frequency textural details from low-frequency illumination signals.

4.3 Software Design Patterns

The system utilizes the **Modular Factory** design pattern. Each restoration stage (ANFIS, LCR, Swin) is implemented as a standalone class with a standardized interface. This "pluggable" design is critical for research and deployment; it allows us to swap in different transformer backbones or dictionary sets without modifying the core inference loop. This modularity ensures that the system is resilient to future technological shifts in any specific sub-module.

4.4 Hardware Topology and Parallelization

The ANFIS-LLFSR system is designed to leverage high-performance GPU resources for parallelized inference.

4.4.1 Workstation Configuration

The experiments and inference benchmarks were conducted on a dual-GPU workstation:

- **GPU**: 2x NVIDIA GeForce RTX 3090 (24GB VRAM each) connected via NVLink.
- **CPU**: AMD Ryzen 9 5950X (16 Cores, 32 Threads).
- **RAM**: 128GB DDR4 3600MHz.

4.4.2 Heterogeneous Parallelism

To achieve near-real-time performance, we utilize a "Heterogeneous Parallelism" approach. The memory-intensive LCR dictionary atoms are stored in GPU 0, while the compute-intensive Swin Transformer attention maps are processed in GPU 1. By utilizing CUDA streams, we overlap the dictionary search (Algorithm ??) with the neural feature modulation (Algorithm ??), resulting in a 22% reduction in total inference latency compared to a single-GPU serial implementation.

4.5 Deployment Interface: The Real-Time Dashboard

To facilitate the practical utility of the model for forensic analysts, we developed a real-time dashboard using FastAPI (backend) and React (frontend).

4.5.1 Asynchronous Inference Pipeline

The dashboard utilizes a producer-consumer architecture. Incoming image streams from surveillance feeds are queued and processed asynchronously. The ANFIS module provides a live "Darkness Heatmap," alerting the operator when a frame requires heavy structural hallucination due to severe photon starvation.

4.5.2 Identity Verification Sentinel

The dashboard integrates a persistent embedding cache. Every restored face is automatically compared against a local database using the ArcFace sentinel. If the similarity score exceeds a predefined threshold (e.g., 0.70), the system flags the identity as a "Forensic Match," providing a confidence score alongside the restored high-resolution image.

Chapter 5

CONCLUSION AND FUTURE WORK

5.1 Summary of Thesis Achievements

This research has successfully bridged the gap between interpretable fuzzy inference and high-capacity neural super-resolution through the development of the ANFIS-LLFSR framework. By integrating a neuro-fuzzy "meta-controller" with locality-constrained hallucination and transformer-based refinement, we have addressed the critical challenge of identity preservation in severe low-light conditions.

5.1.1 Key Technical Contributions

1. **Interpretable Illumination Modulation:** We implemented a 5-layer ANFIS module that provides an explicit, linguistic-driven layer for illumination estimation. This "Fuzzy Router" allows forensic analysts to verify the model's enhancement logic, moving away from the black-box nature of standard CNNs.
2. **Structural Identity Anchoring:** Through the combination of a locality-constrained representation (LCR) dictionary and an ArcFace-guided loss sentinel, we achieved a significant reduction in "generative hallucinations," ensuring that the restored facial features remain faithful to the original subject.
3. **Synergistic Neural Architecture:** The integration of Swin Transformer blocks with SFT modulation and Wavelet-Frequency fusion demonstrated state-of-the-art performance, achieving a PSNR of 30.42 dB and an ArcFace similarity of 0.758 under compound degradation.

5.2 Practical Impact and Possible Beneficiaries

The findings and artifacts produced in this thesis have direct implications for several domains:

- **Forensic Science and Law Enforcement:** Providing a reliable tool for enhancing night-time surveillance footage where identification was previously impossible.
- **Biometric Security Systems:** Improving the robustness of face-unlock and secure entry protocols in unconstrained lighting environments.

- **Autonomous Vehicle Monitoring:** Enhancing driver-facing cameras for night-time drowsiness detection and safety monitoring.

5.3 Limitations and Boundaries

Despite its robustness, the ANFIS-LLFSR system faces specific operational limitations:

- **Computational Latency:** The patch-wise dictionary search in the LCR stage remains the primary bottleneck, limiting the system to near-real-time (1-2 FPS) performance on server-grade GPUs.
- **Extreme Pose Invariance:** While robust to frontal and semi-profile views, the model's accuracy exhibits a decay for extreme side profiles where the symmetry-based priors of the transformer are less effective.

5.4 Future Research Directions

5.4.1 Temporal Consistency in Video Forensics

A promising direction is the extension of this framework to handle video sequences. By incorporating recurrent modules (e.g., ConvLSTM) or temporal attention, the system can leverage information from multiple frames to ensure that facial features remain stable and consistent across a sequence, which is essential for court-admissible evidence.

5.4.2 Diffusion-Based Generative Anchoring

While this work utilized Transformers for refinement, the emergence of Latent Diffusion Models (LDMs) offers a pathway to even higher perceptual realism. Future research will explore "fuzzy-anchored diffusion," where the reverse diffusion process is steered by the ANFIS Darkness Factor to prevent the stochastic generation of false features.

5.4.3 Privacy-Preserving Federated Learning

To improve model robustness across diverse global demographics without compromising data privacy, a federated learning approach could be adopted. This would allow the system to learn from sensitive surveillance data at the edge, ensuring biometric privacy while improving the generalizability of the facial manifold.

5.5 Final Concluding Remarks

The development of ANFIS-LLFSR demonstrates that the synergy between classical fuzzy logic and modern deep learning provides a powerful pathway for solving complex, real-world problems. By anchoring the hallucination process in both global context and local identity, we have produced a system that is not only visually impressive but also technically reliable for critical security applications.

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Chapter 6

CORE COMPUTATIONAL ALGORITHMS

6.1 Core Algorithms

Algorithm 1 ANFIS Hybrid Learning (Forward Pass)

- 1: **Input:** Training data $(\mathbf{x}, y_{target})$
 - 2: **Layer 1:** Compute Gaussian membership degrees $\mu_{i,k}(x_i)$
 - 3: **Layer 2:** Compute firing strengths $w_r = \prod \mu_{i,k}$
 - 4: **Layer 3:** Normalize strengths $\bar{w}_r = w_r / \sum w_j$
 - 5: **Layer 4:** Solve for consequent parameters $\mathbf{P}$ using Recursive Least Squares:
 - 6: $\mathbf{y} = \mathbf{A}\mathbf{P} \implies \mathbf{P}^* = (\mathbf{A}^T \mathbf{A} + \lambda \mathbf{I})^{-1} \mathbf{A}^T \mathbf{y}$
 - 7: **Output:** Consequent parameters $\mathbf{P}^*$
-

Algorithm 2 ANFIS-Guided LCR Patch Hallucination

- 1: **Input:** LR Image I_{LR} , ANFIS Darkness Factor DF
 - 2: **Step 1:** Extract LR patches $\{p_i\}$ with stride s
 - 3: **Step 2:** For each patch p_i :
 - 4: Compute locality weights $d_k = \exp(\|p_i - D_{LR,k}\|^2 / \beta)$
 - 5: Solve LCR coefficients $\alpha^* = (D_{LR}^T D_{LR} + \lambda \text{diag}(d^2))^{-1} D_{LR}^T p_i$
 - 6: Compute affinity $a_k = 1 - |DF - DF_{atom,k}|$
 - 7: Reweight coefficients $\alpha_{rw} = \text{Norm}(\alpha^* \odot a)$
 - 8: Hallucinate HR patch $h_i = D_{HR} \cdot \alpha_{rw}$
 - 9: **Step 3:** Assemble $\{h_i\}$ using overlap averaging
 - 10: **Output:** Hallucinated Face I_{HR}
-

Algorithm 3 Spatial Feature Transform (SFT) Modulation

- 1: **Input:** Deep feature map F , Condition vector $C = [DF, \dots]$
 - 2: **Step 1:** Compute scale parameter $\gamma = \text{MLP}_{\text{scale}}(C)$
 - 3: **Step 2:** Compute shift parameter $\beta = \text{MLP}_{\text{shift}}(C)$
 - 4: **Step 3:** Compute residual gate $g = \text{Sigmoid}(\text{MLP}_{\text{gate}}(C))$
 - 5: **Step 4:** Apply modulation: $F_{out} = F + (F \odot \gamma + \beta) \odot g \cdot \alpha_{enhance}$
 - 6: **Output:** Modulated feature map F_{out}
-

6.1.1 Computational Analysis of Restoration Algorithms

The algorithms presented in this section are designed for maximal GPU utilization. Algorithm 2 (LCR) is a memory-bound operation, as it requires storing the 1024×1024 covariance matrix for each dictionary atom. In contrast, Algorithm 3 (SFT) is a compute-bound operation that scales with the depth of the transformer blocks. In our production environment, we utilize CUDA streams to overlap the dictionary lookup of Algorithm 2 with the feature modulation of Algorithm 3, achieving a 22% reduction in total inference latency.

Chapter 7

EXTENDED MATHEMATICAL PROOFS

7.1 Locality Constrained Representation (LCR) Closed-Form Derivation

The LCR objective function is:

$$\min_{\alpha} \|p - D\alpha\|^2 + \lambda \|d \odot \alpha\|^2 \quad \text{s.t. } \mathbf{1}^T \alpha = 1 \quad (7.1)$$

Expanding the terms:

$$\|p - D\alpha\|^2 = (p - D\alpha)^T (p - D\alpha) = p^T p - 2\alpha^T D^T p + \alpha^T D^T D \alpha \quad (7.2)$$

The penalty term:

$$\|d \odot \alpha\|^2 = \alpha^T \text{diag}(d^2) \alpha \quad (7.3)$$

Combining and adding the Lagrange multiplier μ :

$$L(\alpha, \mu) = \alpha^T (D^T D + \lambda \text{diag}(d^2)) \alpha - 2\alpha^T D^T p + p^T p + \mu(\mathbf{1}^T \alpha - 1) \quad (7.4)$$

Taking the derivative w.r.t. α and setting to zero:

$$\frac{\partial L}{\partial \alpha} = 2(D^T D + \lambda \text{diag}(d^2))\alpha - 2D^T p + \mu \mathbf{1} = 0 \quad (7.5)$$

Solving for α :

$$\alpha = (D^T D + \lambda \text{diag}(d^2))^{-1} (D^T p - \frac{\mu}{2} \mathbf{1}) \quad (7.6)$$

The normalization constraint $\mathbf{1}^T \alpha = 1$ is used to solve for μ , leading to the finalized LCR coefficients used in our pipeline.

Chapter 8

GLOSSARY OF TECHNICAL TERMS

ANFIS Adaptive Neuro-Fuzzy Inference System. A hybrid intelligence model combining fuzzy logic and neural networks.

ArcFace A state-of-the-art additive angular margin loss for deep face recognition.

CBAM Convolutional Block Attention Module. An attention module for deep convolutional neural networks.

LCR Locality-Constrained Representation. A sparse coding technique that incorporates locality constraints for better projection.

Swin Transformer Hierarchical vision transformer using shifted windows.

SFT Spatial Feature Transform. A modulation technique used to condition neural features on external signals.

PSNR Peak Signal-to-Noise Ratio. A metric for image reconstruction quality.

SSIM Structural Similarity Index Measure. A metric for measuring the similarity between two images.

LPIPS Learned Perceptual Image Patch Similarity. A deep-learning based perceptual metric.

Wavelet Fusion A multi-resolution analysis technique for combining information from different frequency bands.

RLS Recursive Least Squares. An adaptive filter algorithm used for parameter optimization.

GD Gradient Descent. An iterative optimization algorithm for finding the local minimum of a function.

SNR Signal-to-Noise Ratio. A measure of the level of a desired signal to the level of background noise.

Forensic Restoration The process of recovering identifiable information from degraded digital evidence for legal purposes.

Chapter 9

LIST OF PUBLICATIONS AND MANUSCRIPTS

The research conducted as part of this B.Tech thesis has led to the following publications and manuscripts in preparation:

1. **Mulla, H.**, Sharma, D. K., Mohiuddin, M. A., and Jain, A. (2024). "ANFIS-LLFSR: A Neuro-Fuzzy Guided Hybrid for Low-Light Face Super-Resolution." *IEEE International Conference on Image Processing (ICIP)*. (Manuscript under review).
2. Sharma, D. K., **Mulla, H.**, and Jain, A. (2023). "Adaptive Illumination Estimation using ANFIS for Night-time Surveillance." *NSUT Tech Journal*. Vol. 12, No. 2, pp. 45-52.
3. Mohiuddin, M. A., and **Mulla, H.** (2024). "Identity-Preserving Generative Facial Priors in Extreme Low-Light Conditions." *CVPR Workshops*. (Manuscript in preparation).

Chapter 10

RESEARCH ARTIFACTS REPOSITORY

The source code, pre-trained models, and experimental datasets are hosted on GitHub for open-access research:

- **GitHub Repository:** <https://github.com/huzaifamulla/ANFIS-LLFSR>
- **Model Zoo:** Includes weights for SwinFuzzyLCR-Base and SwinFuzzyLCR-Large.
- **Dataset:** Synthetic CelebA-HQ degradation scripts and LFW test results.

10.1 Wavelet-Frequency Fusion Logic

Algorithm 4 Wavelet-Frequency Fusion for High-Resolution Recovery

- 1: **Input:** Refined feature map F_{ref} , Original input I_{LR}
 - 2: **Step 1:** Perform 2-level Discrete Wavelet Transform (DWT) on F_{ref} :
 - 3: $(LL, LH, HL, HH) = \text{DWT}(F_{ref})$
 - 4: **Step 2:** Enhance high-frequency sub-bands (LH, HL, HH) using residual blocks:
 - 5: $LH' = \text{ResBlock}(LH)$
 - 6: $HL' = \text{ResBlock}(HL)$
 - 7: $HH' = \text{ResBlock}(HH)$
 - 8: **Step 3:** Compute frequency-domain attention map M_{freq} using FFT on LL :
 - 9: $M_{freq} = \text{Sigmoid}(\text{Conv}(\text{FFT}(LL)))$
 - 10: **Step 4:** Modulate low-frequency band: $LL' = LL \odot M_{freq}$
 - 11: **Step 5:** Perform Inverse DWT to reconstruct features:
 - 12: $F_{fused} = \text{IDWT}(LL', LH', HL', HH')$
 - 13: **Output:** Frequency-fused features F_{fused}
-

Chapter 11

FORENSIC DEPLOYMENT FLOWCHARTS



Figure 11.1: Hardware topology for a distributed forensic restoration system, showing the load balancer, GPU workers, and the persistent ArcFace identity database.